## IV. Match the following

**Match: I**

| | |
|---|---|
| a. BARC | Kalpakkam |
| b. India's first atomic power station | Apsara |
| c. IGCAR | Mumbai |
| d. First nuclear reactor in India | Tarapur |

**Match: II**

| | |
|---|---|
| a. Fuel | lead |
| b. Moderator | heavy water |
| c. Control rods | cadmium rods |
| d. Shield | uranium |

**Match: III**

| | |
|---|---|
| a. Soddy Fajan | Natural radioactivity |
| b. Irene Curie | Displacement law |
| c. Henry Bequerel | Mass energy equivalence |
| d. Albert Einstein | Artificial Radioactivity |

**Match: IV**

| | |
|---|---|
| a. Uncontrolled fission reaction | Hydrogen Bomb |
| b. Fertile material | Nuclear Reactor |
| c. Controlled fission reaction | Breeder reactor |
| d. Fusion reaction | Atom bomb |

**Match: V**

| | |
|---|---|
| a. Fe - 59 | Age of fossil |
| b. I - 131 | Function of Heart |
| c. Na - 24 | Leukemia |
| d. C - 14 | Thyroid disease |

## V. Arrange the following in the correct sequence:

1. **Arrange in descending order, on the basis of their penetration power**

   Alpha rays, beta rays, gamma rays, cosmic rays

2. **Arrange the following in the chronological order of discovery**

   Nuclear reactor, radioactivity, artificial radioactivity, discovery of radium.

## VI. Use the analogy to fill in the blank

1. Spontaneous process : Natural Radioactivity, Induced process : _____
2. Nuclear Fusion : Extreme temperature, Nuclear Fission : _____
3. Increasing crops : Radio phosphorous, Effective functioning of heart : _____
4. Deflected by electric field : α ray, Null Deflection : _____

## VII. Numerical problems:

1. $_{88}Ra^{226}$ experiences three **α** - decay. Find the number of neutrons in the daughter element.
2. A cobalt specimen emits induced radiation of 75.6 millicurie per second. Convert this disintegration in to becquerel (one curie = $3.7 \times 10^{10}$ Bq)

## VIII. Assertion and reason type questions:

**Mark the correct choice as**

(a) If both the assertion and the reason are true and the reason is the correct explanation of the assertion.

(b) If both the assertion and the reason are true, but the reason is not the correct explanation of the assertion.

(c) Assertion is true, but the reason is false.

(d) Assertion is false, but the reason is true.

1. **Assertion:** A neutron impinging on $U^{235}$, splits it to produce Barium and Krypton.
   **Reason**: U - 235 is a fissile material.
2. **Assertion**: In a β - decay, the neutron number decreases by one.
   **Reason**: In β - decay atomic number increases by one.
3. **Assertion**: Extreme temperature is necessary to execute nuclear fusion.
   **Reason**: In a nuclear fusion, the nuclei of the reactants combine releasing high energy.



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